

(a) $\Rightarrow$ (b): Suppose statement (a) holds. Further suppose that $0 \notin \text{conv}(\mathcal{M})$.

Since $\text{conv}(\mathcal{M})$ is compact, by Corollary 5.11, there is one $\beta_* \in \text{conv}(\mathcal{M})$, $\beta_* \neq 0$, of minimal Euclidean norm in $\text{conv}(\mathcal{M})$. This minimum β_* , viewed as a best approximation from $\text{conv}(\mathcal{M})$ to the origin with respect to $\|\cdot\|_2$, is characterized by

$$(\beta_* - 0, x - \beta_*) \geq 0 \quad \text{for all } x \in \text{conv}(\mathcal{M})$$

according to the Kolmogorov theorem, Corollary 3.55, in combination with the Gâteaux derivative for Euclidean norms in Theorem 3.62. But this condition is equivalent to

$$\beta_*^T x = (\beta_*, x) \geq (\beta_*, \beta_*) = \|\beta_*\|_2^2 > 0 \quad \text{for all } x \in \text{conv}(\mathcal{M}).$$

But this is in contradiction to our assumption in (a). ■

Remark 5.13. The equivalence statement (a) in Theorem 5.12 says that the Euclidean space $\mathbb{R}^d$ cannot be split by a separating hyperplane through the origin into two half-spaces, such that $\mathcal{M}$ is entirely contained in one of the two half-spaces. □

5.2 Strongly Unique Best Approximations

Now we wish to further develop the characterizations for best approximations from Sections 3.3 and 3.4 for the special case of the Chebyshev norm $\|\cdot\|_\infty$. In the following discussion, $\{s_1, \dots, s_n\} \subset \mathcal{S}$, for $n \in \mathbb{N}$, denotes a basis for the finite-dimensional approximation space $\mathcal{S} \subset \mathcal{C}(\Omega)$. To characterize a best approximation $s^* \in \mathcal{S}$ to some $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$, we work with the compact point set

$$\mathcal{M}_{s^*-f} = \{(s^* - f)(x)(s_1(x), \dots, s_n(x))^T \mid x \in E_{s^*-f}\} \subset \mathbb{R}^n,$$

where we can immediately draw the following conclusion from Theorem 5.12.

Corollary 5.14. *For $s^* \in \mathcal{S}$ the following statements are equivalent.*

- (a) s^* is a best approximation to $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$.
- (b) $0 \in \text{conv}(\mathcal{M}_{s^*-f})$.

Proof. In this proof, we use the notation

$$s_\beta = \sum_{j=1}^n \beta_j s_j \in \mathcal{S} \quad \text{for } \beta = (\beta_1, \dots, \beta_n)^T \in \mathbb{R}^n. \quad (5.10)$$

(b) $\Rightarrow$ (a): Let $0 \in \text{conv}(\mathcal{M}_{s^*-f})$. Suppose $s^* \in \mathcal{S}$ is *not* a best approximation to f . Then there is one $\beta = (\beta_1, \dots, \beta_n)^T \in \mathbb{R}^n \setminus \{0\}$ satisfying

$$\|s^* - f - s_\beta\|_\infty < \|s^* - f\|_\infty$$

In this case, we have

$$|(s^* - f)(x) - s_\beta(x)|^2 < |(s^* - f)(x)|^2 \quad \text{for all } x \in E_{s^*-f}.$$

But this is equivalent to

$$|(s^* - f)(x)|^2 - 2(s^* - f)(x)s_\beta(x) + s_\beta^2(x) < |(s^* - f)(x)|^2$$

for all $x \in E_{s^*-f}$, so that

$$(s^* - f)(x)s_\beta(x) > \frac{1}{2}s_\beta^2(x) \geq 0 \quad \text{for all } x \in E_{s^*-f},$$

i.e.,

$$\beta^T(s^* - f)(x)(s_1(x), \dots, s_n(x))^T > 0 \quad \text{for all } x \in E_{s^*-f}.$$

By the equivalence statements in Theorem 5.12, we see that the origin is in this case *not* contained in the convex hull $\text{conv}(\mathcal{M}_{s^*-f})$. But this is in contradiction to statement (b).

(a) $\Rightarrow$ (b): Let s^* be a best approximation to f . Suppose $0 \notin \text{conv}(\mathcal{M}_{s^*-f})$. Due to Theorem 5.12, there is one $\beta = (\beta_1, \dots, \beta_n)^T \in \mathbb{R}^n \setminus \{0\}$ satisfying $\beta^T u > 0$, or, $-\beta^T u < 0$, for all $u \in \mathcal{M}_{s^*-f}$. But this is equivalent to

$$(s^* - f)(x)s_{-\beta}(x) < 0 \quad \text{for all } x \in E_{s^*-f},$$

i.e., $s^* - f$ and $s_{-\beta}$ have opposite signs on E_{s^*-f} , whereby

$$\text{sgn}((s^* - f)(x))s_{-\beta}(x) < 0 \quad \text{for all } x \in E_{s^*-f}.$$

In particular (by using the compactness of E_{s^*-f}), we have

$$\max_{x \in E_{s^*-f}} \text{sgn}((s^* - f)(x))s_{-\beta}(x) < 0.$$

But this is, due to the Kolmogorov criterion, Theorem 5.1, in contradiction to the optimality of s^* in (a), ■

Corollary 5.14 yields an important result concerning the characterization of best approximations.

Corollary 5.15. *For $s^* \in \mathcal{S}$ the following statements are equivalent.*

- (a) s^* is a best approximation to $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$.
- (b) There are $m \leq n + 1$
 - pairwise distinct extremal points $x_1, \dots, x_m \in E_{s^*-f}$
 - signs $\varepsilon_j = \text{sgn}((s^* - f)(x_j))$, for $j = 1, \dots, m$,
 - coefficients $\lambda = (\lambda_1, \dots, \lambda_m)^T \in \Lambda_m$ with $\lambda_j > 0$ for all $1 \leq j \leq m$, satisfying

$$\varphi(s) := \sum_{j=1}^m \lambda_j \varepsilon_j s(x_j) = 0 \quad \text{for all } s \in \mathcal{S}. \quad (5.11)$$

Proof. (a) $\Rightarrow$ (b): Let s^* be a best approximation to f .

Then we have $0 \in \text{conv}(\mathcal{M}_{s^*-f})$ by Corollary 5.14. According to the Carathéodory theorem, Theorem 5.10, there are $m \leq n+1$ extremal points $x_1, \dots, x_m \in E_{s^*-f}$ and coefficients $\lambda = (\lambda_1, \dots, \lambda_m)^T \in \Lambda_m$ satisfying

$$0 = \sum_{j=1}^m \lambda_j ((s^* - f)(x_j)) s_k(x_j) = \sum_{j=1}^m \lambda_j \varepsilon_j \|s^* - f\|_{\infty} s_k(x_j) = \sum_{j=1}^m \lambda_j \varepsilon_j s_k(x_j)$$

for all basis elements $s_k \in \mathcal{S}$, $k = 1, \dots, n$.

(b) $\Rightarrow$ (a): Under the assumption in (b), we have $0 \in \text{conv}(\mathcal{M}_{s^*-f})$, whereby s^* is a best approximation to f , due to Corollary 5.14. ■

Remark 5.16. In statement (b) of Corollary 5.15 the alternation condition

$$\varepsilon_j \cdot \varepsilon_{j+1} = -1 \quad \text{for } j = 1, \dots, m-1$$

is *not* necessarily satisfied. In Corollary 5.4, we considered the special case of polynomial approximation by $\mathcal{S} = \mathcal{P}_{n-1} \subset \mathcal{C}[a, b]$. In that case, the alternation condition (5.3) is satisfied with *at least* $n+1$ extremal points. But in Corollary 5.15 only *at most* $n+1$ extremal points are allowed. □

In the following discussion, we will see how the characterizations of Corollary 5.4 and Corollary 5.15 can be combined. To this end, the following result is of central importance, whereby we can even prove *strong* uniqueness for best approximations.

Theorem 5.17. For $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$, let $s^* \in \mathcal{S}$ be a best approximation to f . Moreover, suppose that $\varphi : \mathcal{C}(\Omega) \rightarrow \mathbb{R}$ is a linear functional of the form

$$\varphi(u) = \sum_{k=1}^m \lambda_k \varepsilon_k u(x_k) \quad \text{for } u \in \mathcal{C}(\Omega) \quad (5.12)$$

satisfying the dual characterization (5.11) of Corollary 5.15 for a point set $X = \{x_1, \dots, x_m\} \subset E_{s^*-f}$, where $2 \leq m \leq n+1$. Then, we have for any $s \in \mathcal{S}$ the estimates

$$\|s - f\|_{\infty} \geq \|s - f\|_{\infty, X} \geq \|s^* - f\|_{\infty} + \frac{\lambda_{\min}}{1 - \lambda_{\min}} \|s^* - s\|_{\infty, X}, \quad (5.13)$$

where $\lambda_{\min} := \min_{1 \leq j \leq m} \lambda_j > 0$.

Proof. Suppose $s \in \mathcal{S}$. Then, the first estimate in (5.13) is trivial. To show the second estimate in (5.13), we use the ingredients $\varepsilon \in \{\pm 1\}^m$, $\lambda \in \Lambda_m$, and $X \subset E_{s^*-f}$ for the functional φ in (5.12) from the dual characterization of Corollary 5.15. Note that we have

$$\|s - f\|_{\infty, X} \geq \varepsilon_j (s - f)(x_j) = \varepsilon_j (s - s^*)(x_j) + \varepsilon_j (s^* - f)(x_j)$$

and, moreover, $\varepsilon_j(s^* - f)(x_j) = \|s^* - f\|_\infty$, for all $j = 1, \dots, m$, so that

$$\|s - f\|_{\infty, X} \geq \|s^* - f\|_\infty + \varepsilon_j(s - s^*)(x_j) \quad \text{for all } 1 \leq j \leq m. \quad (5.14)$$

Since $m \geq 2$, we have $\lambda_{\min} \in (0, 1/2]$ and so $\lambda_{\min}/(1 - \lambda_{\min}) \in (0, 1]$.

Now let $x_{j^*} \in X$ be a point satisfying $|(s - s^*)(x_{j^*})| = \|s - s^*\|_{\infty, X}$. If $\varepsilon_j(s - s^*)(x_{j^*}) = \|s - s^*\|_{\infty, X}$, then the second estimate in (5.13) is satisfied, with $\lambda_{\min}/(1 - \lambda_{\min}) \leq 1$. Otherwise, we have $\varepsilon_j(s - s^*)(x_{j^*}) = -\|s - s^*\|_{\infty, X}$, whereby with $\varphi(s - s^*) = 0$ the estimate

$$\lambda_{j^*} \|s - s^*\|_{\infty, X} = \sum_{\substack{k=1 \\ k \neq j^*}}^m \lambda_k \varepsilon_k(s - s^*)(x_k) \leq (1 - \lambda_{j^*}) \max_{k \neq j^*} \varepsilon_k(s - s^*)(x_k) \quad (5.15)$$

follows. Now then, for $k^* \in \{1, \dots, m\} \setminus \{j^*\}$ satisfying

$$\varepsilon_{k^*}(s - s^*)(x_{k^*}) = \max_{k \neq j^*} \varepsilon_k(s - s^*)(x_k)$$

we find, due to (5.15), the estimate

$$\frac{\lambda_{\min}}{1 - \lambda_{\min}} \|s^* - s\|_{\infty, X} \leq \frac{\lambda_{j^*}}{1 - \lambda_{j^*}} \|s^* - s\|_{\infty, X} \leq \varepsilon_{k^*}(s - s^*)(x_{k^*}),$$

which implies, in combination with (5.14), the second estimate in (5.13). ■

Note that for any best approximation $s^* \in \mathcal{S}$ to $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$, the estimates in (5.13) yield the inequality

$$\|s - f\|_\infty - \|s^* - f\|_\infty \geq \frac{\lambda_{\min}}{1 - \lambda_{\min}} \|s - s^*\|_{\infty, X} \quad \text{for all } s \in \mathcal{S}. \quad (5.16)$$

Given this result, we can further analyze the question for the (strong) uniqueness of best approximations to $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$. To this end, we first take note of the following simple observation.

Remark 5.18. Let $s^* \in \mathcal{S}$ be a best approximation to $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$. Then, for any other best approximation $s^{**} \in \mathcal{S}$ to $f \in \mathcal{C}(\Omega)$ we have

$$0 = \|s^{**} - f\|_\infty - \|s^* - f\|_\infty \geq \frac{\lambda_{\min}}{1 - \lambda_{\min}} \|s^{**} - s^*\|_{\infty, X}$$

by (5.16), and this implies, for $\lambda_{\min} \in (0, 1)$, the identity

$$\|s^{**} - s^*\|_{\infty, X} = 0.$$

In conclusion, all best approximations to f must coincide on X . Now if $\|\cdot\|_{\infty, X}$ is a *norm* on $\mathcal{S}$, then s^* will be the *unique* best approximation to f . □

In the following Section 5.3, we develop suitable conditions on $\mathcal{S} \subset \mathcal{C}(\Omega)$, under which we can guarantee the uniqueness of best approximations. In our developments the definiteness of $\|\cdot\|_{\infty, X}$ plays an important role. As we can show already now, the definiteness of $\|\cdot\|_{\infty, X}$ guarantees the *strong* uniqueness of a best approximation $s^* \in \mathcal{S}$ to $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$.

Theorem 5.19. *Under the assumptions of Theorem 5.17, let $\|\cdot\|_{\infty, X}$ be a norm on $\mathcal{S}$. Then there exists for any $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$ a strongly unique best approximation $s^* \in \mathcal{S}$ to f .*

Proof. The approximation space $\mathcal{S} \subset \mathcal{C}(\Omega)$ is finite-dimensional. Therefore, there exists for any $f \in \mathcal{C}(\Omega)$ a best approximation $s^* \in \mathcal{S}$ to f , according to Corollary 3.8. Moreover, all norms on $\mathcal{S}$ are equivalent. In particular, the two norms $\|\cdot\|_{\infty}$ and $\|\cdot\|_{\infty, X}$ are on $\mathcal{S}$ equivalent, so that there is a constant $\beta > 0$ satisfying

$$\|s\|_{\infty, X} \geq \beta \|s\|_{\infty} \quad \text{for all } s \in \mathcal{S}. \quad (5.17)$$

By (5.13), the best approximation s^* to f is strongly unique, since

$$\|s - f\|_{\infty} - \|s^* - f\|_{\infty} \geq \frac{\lambda_{\min}}{1 - \lambda_{\min}} \|s - s^*\|_{\infty, X} \geq \alpha \|s - s^*\|_{\infty} \quad \text{for all } s \in \mathcal{S},$$

where $\alpha = \beta \lambda_{\min} / (1 - \lambda_{\min}) > 0$. ■

Before we continue our analysis, we first discuss two examples.

Example 5.20. Let $\mathcal{F} = \mathcal{C}[-1, 1]$, $\mathcal{S} = \mathcal{P}_1 \subset \mathcal{F}$ and $f(x) = x^2$. Then, $c^* \equiv 1/2$ is according to Corollary 5.2 the unique best approximation to f from $\mathcal{P}_0$. Since f is even, the unique best approximation $p_1^* \in \mathcal{P}_1$ to f from $\mathcal{P}_1$ is also even, due to Corollary 5.5. In this case, p_1^* is necessarily constant, and so c^* is also the unique best approximation to f from $\mathcal{P}_1$. Moreover, the error function $c^* - f$ has on the interval $[-1, 1]$ exactly three extremal points $X = \{x_1, x_2, x_3\} = \{-1, 0, 1\}$, where the alternation conditions are satisfied,

$$c^* - f(x_j) = (-1)^j \|c^* - f\|_{\infty} = (-1)^j \cdot \frac{1}{2} \quad \text{for } j = 1, 2, 3.$$

For $\lambda_1 = 1/4$, $\lambda_2 = 1/2$, $\lambda_3 = 1/4$ and $\varepsilon_j = (-1)^j$, for $j = 1, 2, 3$, we have

$$\sum_{j=1}^3 \lambda_j \varepsilon_j p(x_j) = 0 \quad \text{for all } p \in \mathcal{P}_1.$$

Moreover, $\|\cdot\|_{\infty, X}$ is a norm on $\mathcal{P}_1$. According to Theorem 5.19, the constant c^* is the strongly unique best approximation to f from $\mathcal{P}_1$. By $\lambda_{\min} = \min_{1 \leq j \leq 3} \lambda_j = 1/4$ we get, like in the proof of Theorem 5.19 (with $\beta = 1$), the estimate

$$\|p - f\|_{\infty} - \|c^* - f\|_{\infty} \geq \frac{\lambda_{\min}}{1 - \lambda_{\min}} \|p - c^*\|_{\infty, X} = \frac{1}{3} \|p - c^*\|_{\infty} \quad \text{for all } p \in \mathcal{P}_1$$

for the strong uniqueness of c^* with the constant $\alpha = 1/3$. ◇

For further illustration, we make the following link to Example 5.7.

Example 5.21. Let $\mathcal{F} = \mathcal{C}[-1, 1]$, $\mathcal{S} = \mathcal{P}_2 \subset \mathcal{F}$ and $f(x) = |x|$. From Example 5.7 the function $p_2^*(x) = 1/8 + x^2$ is the unique best approximation to f from $\mathcal{P}_2$ with extremal point set $E_{p_2^*-f} = \{0, \pm 1/2, \pm 1\}$.

For the dual characterization of the best approximation $p_2^* \in \mathcal{P}_2$ to f , we seek, according to Corollary 5.15, a set $X \subset E_{p_2^*-f}$ of extremal points, where $2 \leq m = |X| \leq \dim(\mathcal{P}_2) + 1 = 4$, signs $\varepsilon_j = \text{sgn}((p_2^* - f)(x_j))$, $1 \leq j \leq m$, and coefficients $\lambda = (\lambda_1, \dots, \lambda_m) \in \Lambda_m$ satisfying

$$\sum_{j=1}^m \lambda_j \varepsilon_j p(x_j) = 0 \quad \text{for all } p \in \mathcal{P}_2. \quad (5.18)$$

This results in $\dim(\mathcal{P}_2) = 3$ linear equations. Together with

$$\lambda_1 + \dots + \lambda_m = 1 \quad (5.19)$$

we get a total number of *four* linear equation conditions for $\lambda \in \Lambda_m$. Therefore, we let $m = 4$ and, moreover, we take $X = \{-1/2, 0, 1/2, 1\} \subset E_{p_2^*-f}$ with signs $\varepsilon = (-1, 1, -1, 1)$. In this way, we reformulate (5.18) as follows.

$$-\lambda_1 p(-1/2) + \lambda_2 p(0) - \lambda_3 p(1/2) + \lambda_4 p(1) = 0 \quad \text{for all } p \in \mathcal{P}_2. \quad (5.20)$$

We pose the conditions from (5.20) to the three elements of the monomial basis $\{1, x, x^2\}$ of $\mathcal{P}_2$. For $p \equiv 1$ we get $-\lambda_1 + \lambda_2 - \lambda_3 + \lambda_4 = 0$, whereby from (5.19) we get

$$\lambda_2 + \lambda_4 = 1/2 \quad \text{and} \quad \lambda_1 + \lambda_3 = 1/2. \quad (5.21)$$

For $p(x) = x$ and $p(x) = x^2$, we get by (5.20) the conditions

$$\lambda_4 = \frac{1}{2}(\lambda_3 - \lambda_1) \quad \text{and} \quad \lambda_4 = \frac{1}{4}(\lambda_1 + \lambda_3). \quad (5.22)$$

Then, (5.21) implies $\lambda_4 = 1/8$ and moreover $\lambda_2 = 3/8$. From (5.21) and (5.22) we finally compute $\lambda_3 = 3/8$ and $\lambda_1 = 1/8$. Therefore,

$$\lambda_{\min} = \frac{1}{8} \quad \text{and} \quad \frac{\lambda_{\min}}{1 - \lambda_{\min}} = \frac{1}{7}.$$

The characterization (5.13) in Theorem 5.17 implies the estimate

$$\|p - f\|_{\infty} - \|p_2^* - f\|_{\infty} \geq \frac{1}{7} \|p - p_2^*\|_{\infty, X} \quad \text{for all } p \in \mathcal{P}_2. \quad (5.23)$$

Next, we show the strong uniqueness of p_2^* , where we use Theorem 5.19. To this end, note that $\|\cdot\|_{\infty, X}$ is a norm on $\mathcal{P}_2$. Therefore, it remains to determine an equivalence constant $\beta > 0$, like in (5.17), satisfying

$$\|p\|_{\infty, X} \geq \beta \|p\|_{\infty} \quad \text{for all } p \in \mathcal{P}_2. \quad (5.24)$$

We choose for $p \in \mathcal{P}_2$ the monomial representation $p(x) = a_0 + a_1x + a_2x^2$. By evaluation of p on the point set $X = \{-1/2, 0, 1/2, 1\}$, we get

$$a_0 = p(0), \quad a_1 = p(1/2) - p(-1/2), \quad a_2 = p(1) - p(0) - p(1/2) + p(-1/2),$$

and therefore the (rough) estimate

$$\|p\|_{\infty} \leq |a_0| + |a_1| + |a_2| \leq \|p\|_{\infty, X} + 2\|p\|_{\infty, X} + 4\|p\|_{\infty, X} = 7\|p\|_{\infty, X}$$

for all $p \in \mathcal{P}_2$, whereby (5.24) holds for $\beta = 1/7$. Together with (5.23), this finally yields the sought estimate

$$\|p - f\|_{\infty} - \|p_2^* - f\|_{\infty} \geq \frac{1}{7} \|p - p_2^*\|_{\infty, X} \geq \frac{1}{49} \|p - p_2^*\|_{\infty} \quad \text{for all } p \in \mathcal{P}_2.$$

Therefore, p_2^* is the strongly unique best approximation to f . $\diamond$

5.3 Haar Spaces

In this section, we develop sufficient conditions for the approximation space $\mathcal{S} \subset \mathcal{C}(\Omega)$ under which a best approximation $s^* \in \mathcal{S}$ to $f \in \mathcal{C}(\Omega) \setminus \mathcal{S}$ is strongly unique. To this end, we can rely on the result of Theorem 5.19, according to which we need to ensure the definiteness of $\|\cdot\|_{\infty, X}$ on $\mathcal{S}$ for any $X \subset E_{s^* - f}$.

We continue to use the assumptions and notations from the previous section, where $(s_1, \dots, s_n) \in \mathcal{S}^n$, for $n \in \mathbb{N}$, denotes an *ordered* basis of a finite-dimensional linear approximation space $\mathcal{S} \subset \mathcal{C}(\Omega)$. By the introduction of *Haar*³ spaces we specialize our assumptions on $\mathcal{S}$ and $(s_1, \dots, s_n)$ as follows.

Definition 5.22. A linear space $\mathcal{S} \subset \mathcal{C}(\Omega)$ with $\dim(\mathcal{S}) = n < \infty$ is called a *Haar space* of dimension $n \in \mathbb{N}$ on Ω , if any $s \in \mathcal{S} \setminus \{0\}$ has at most $n - 1$ zeros on Ω . A basis $\mathcal{H} = (s_1, \dots, s_n) \in \mathcal{S}^n$ for a Haar space $\mathcal{S}$ on Ω is called a *Haar system* on Ω . $\bigcirc$

In Haar spaces $\mathcal{S}$ of dimension $n \in \mathbb{N}$, we can solve interpolation problems for a discrete set $X \subset \Omega$ containing $|X| = n$ pairwise distinct points. In this case, $\|\cdot\|_{\infty, X}$ is a norm on $\mathcal{S}$, and so a solution of the interpolation problem is unique. We can further characterize Haar spaces as follows.

³ ALFRÉD HAAR (1885-1933), Hungarian mathematician